

An Encrypted Traffic Classification Framework Based on Convolutional Neural Networks and Stacked Autoencoders

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Abstract—In recent years, deep learning-based encrypted traffic classification has proven to be effective; especially, using neural networks to extract features from raw traffic to classify encrypted traffic. However, most of the neural networks need a fixed-sized input, so that the raw traffic need to be trimmed. This will cause the loss of some information; for example, we do not know the number of packets in a session. To solve these problems, a framework, which implements both a convolutional neural network (CNN) and a stacked autoencoder (SAE), is proposed in this paper. This framework uses a CNN to extract high-level features from raw network traffic and uses an SAE to encode the 26 statistical features calculated by raw traffic directly. The statistical features can be used to supplement the information loss due to trimming. After that, the outputs from the CNN and the encoder in SAE are combined into new high-level features; these new features include the information from the trimmed raw traffic and statistical features. Finally, these new high-level features are used to classify encrypted traffic. “ISCX VPNnonVPN” traffic dataset is used to demonstrate the feasibility of this framework. The framework proposed in this paper can improve the performance of encrypted traffic classification; it achieves an f1-score of 0.98. Furthermore, new high-level features, which generated by combining the features extracted from a convolutional neural network and a stacked autoencoder, can represent different classes of traffic well. More importantly, this work is unique in the encrypted traffic classification field, for it is the first time to use both raw traffic and statistical features as the input of the model.

Keywords—encrypted traffic classification; stacked autoencoder; convolutional neural network;

I. INTRODUCTION

The recent development of the Internet has made various applications appear, and the amount of traffic generated by these applications is increasing rapidly. A large amount of traffic will make it difficult for the networking industry to ensure the network quality of service (QoS). To cope with the significant traffic increases, traffic classification technologies have been receiving increased attention in recent years. More and more research organizations are making efforts to research on traffic classification and develop new classification technologies [1]. However, as people realize the importance of privacy, encrypted traffic is becoming more and more common in network traffic, which

brings significant challenges to traditional traffic classification. Therefore, some technologies about encrypted traffic classification based on machine learning or deep learning have been proposed [2].

Some approaches for encrypted traffic classification are statistical-based methods. These methods have an assumption that the statistical features of different types of traffic are unique. Draper-Gil et al. used time-related features like the duration of the flow, the time between two packets sent forward direction, and the time between two packets sent in either direction, etc. to classify the encrypted traffic. They achieve 90% accuracy in classifying 12 classes of traffic when using the C4.5 decision tree algorithms [3]. The main drawback of these methods is that the features need to be extracted by experts; it is time-consuming and expensive. The deep neural network can automatically extract features from hierarchical data, and the network traffic flow is hierarchical. Therefore, some approaches based on the deep neural network are proposed. Wang et al. proposed an end-to-end method of encrypted traffic classification [4]. They use a one-dimensional convolutional neural network that automatically extracts features from raw traffic and has an excellent performance on classification.

However, there are still some problems when using deep neural networks to classify encrypted network traffic. Most of the neural networks, like convolutional neural networks, need a fixed-sized input; therefore, the raw network traffic needs to be trimmed first. For example, if the traffic file's size is larger than N bytes, then we trim the file to N bytes. This will cause information loss after N bytes. More importantly, some information about the overall structure of traffic is lost. For example, we will not know the length of the entire traffic and the number of packets in a session, which will affect the classification accuracy.

For the sake of solving these disadvantages, an encrypted traffic classification framework based on convolutional neural networks (CNN) and stacked autoencoders (SAE) is proposed. We use convolutional neural networks to extract features directly from raw encrypted traffic. At the same time, we extract statistical features from raw traffic and use stacked autoencoders to encode these features to high-level features. The statistical features are used to compensate for the information loss caused by traffic trimming. The high-level features extracted from the convolutional neural

network and encoded by the encoder in stacked autoencoder are combined to form new high-level features. The newly generated features can contain information from both trimmed raw traffic and statistical features. Finally, these new high-level features are used for encrypted traffic classification. “ISCX VPN-nonVPN” traffic dataset [3] is used to demonstrate the feasibility of this framework. The results show that this framework has a good performance on classification; it achieves an f1-score of 0.98 in traffic classification. Furthermore, the new high-level features generated by this framework can represent different kinds of network traffic well.

The major contributions of this paper are as follows:

- We propose a framework that employs a convolutional neural network and a stacked autoencoder at the same time. The convolutional neural network is used to extract high-level features from raw traffic, and the stacked autoencoder is used to encode the statistical features.
- The raw traffic needs to be trimmed before it can be input to the convolutional neural network, which will cause information loss, such as how many packets are in a session. In this framework, statistical features are used to supplement some information of the raw traffic.
- This framework can generate new high-level features by combining the features extracted from the convolutional neural network and the stacked autoencoder. We use the new high-level features to classify encrypted traffic and achieve an excellent performance.
- The new high-level features can also represent different classes of network traffic well.
- This work is unique in the encrypted traffic classification field, for it is the first time to use both raw traffic and statistical features as the input of the model at the same time.

The rest of the paper is organized as follows. Related works are discussed in section II. The background in this work is introduced in section III, including the stacked autoencoder and convolutional neural network. Section IV proposes the framework in this paper, including two parts, data preprocess and model’s architecture. Section V presents the experiments carried out using the “ISCX VPN-nonVPN” traffic dataset and shows detailed results. Experiment setup and evaluation metrics are also introduced in this section. Finally, section VI concludes the paper and discusses the future direction.

II. RELATED WORK

Traffic classification technology has developed significantly for many years. The initial traffic classification is based on the port number; this method is considered the easiest and fastest [5]. However, when more and more new applications appear, and some of them use standard registered port numbers to disguise their traffic, the accuracy of classification based on ports gradually decreases. Therefore, a new technology based on payload appears. This

method, also known as deep packet inspection (DPI), uses the pre-defined patterns to inspect the payload in the application layer of packets [6]. However, due to the emergence of encrypted network traffic, the payload-based method’s accuracy also declines. As a result, a new method that is based on statistical features has gradually appeared. This method can handle both encrypted and unencrypted traffic. Draper-Gil et al. used this method to achieve superior performance in encrypted traffic classification [3]. Nevertheless, this method’s main problem is that all the statistical features need to be extracted by experts.

In recent years, deep learning methods have shown state-of-the-art performances in encrypted traffic classification. One of its main advantages is that it can automatically extract features from the raw network traffic. Wang et al. proposed a new system that integrates feature extraction, feature selection, and classifier into an end-to-end framework based on the convolution neural network [4]. Song et al. used a text convolutional neural network to extract critical features for traffic classification [7]. Shapira et al. transformed the basic flow data into a picture and then used image classification deep learning techniques to classify encrypted traffic [8]. Guo et al. utilized the convolutional autoencoder and the convolutional neural network to identify real-time VPN traffic [9]. Lotfollahi et al. proposed a “Deep Packet” scheme, which integrates both feature extraction and classification phases in one system. Moreover, “Deep Packet” can handle both traffic classification and application identification [10].

III. BACKGROUND

A. Stacked Autoencoder

An autoencoder (AE) is an unsupervised neural network that learns to minimize the difference between input data and output data. There are two parts of an AE, an encoder and a decoder. The encoder maps the original data to the code (usually, the dimension of the code is less than the original data). The decoder tries to map the code to the original input. One of the applications of AE is dimensionality reduction [11]. Consider an input $x \in R^n$; the objective of the AE is defined to be $y = x$. The AE tries to learn an objective function $F_{W,b}(x) \approx y$, where the W is the weights of the whole neural network, and the b is the network’s biases. The general loss function of AE is shown in (1). The reconstruction loss in (1) is defined as the L_p distance. Then gradient descent can be used for fine-tuning the weights and biases in AE.

$$\mathbb{L}(W, b) = \min \|x - F_{W,b}(x)\|_p \quad (1)$$

In practice, in order to obtain better results, stacked autoencoder (SAE) is proposed [12]. An SAE consists of several autoencoders where the output of each one is connected to the input of the successive one. The training steps of SAE mainly include the following two steps:

- The parameters of other layers are fixed when training one layer. The output from the previous

layer is used as an input for the next layer, and this will continue until the training is completed.

- After training all the layers, the backpropagation algorithm is used to minimize the reconstruction error. The weights in all the layers will be adjusted.

In this work, SAE is used to deal with the statistical features from the network traffic. We will use SAE to encode statistical features and get high-level features.

B. Convolutional Neural Network

The convolutional neural network (CNN) is another deep neural network, most commonly implemented to analyze images. The architecture of CNN is inspired by the visual cortex in the brain. Individual neurons respond to stimuli only in restricted areas of the visual field, and all these neurons can cover the entire visible area [13], [14]. The convolutional layer is the most fundamental component of CNN. Now consider the input of a convolutional layer is a matrix $X_{N \times N}$ and the convolution kernel is $\mathcal{W}_{m \times m}$. The stride is 1, and the padding is 0. The output of this layer is z of size $(N - m + 1) \times (N - m + 1)$, and (2) is the way to compute z .

$$z_{ij} = f \left(\sum_{a=0}^{m-1} \sum_{b=0}^{m-1} \mathcal{W}_{ab} X_{(i+a)(j+b)} \right), \quad (2)$$

where f is non-linear activations such as Sigmoid, rectified linear activation unit (ReLU), or Leaky ReLU [15]. There is another basic block in CNN, which is the pooling layer, typically max pooling or average pooling. The pooling layer can reduce the size of the output, thereby further reduce the computational complexity.

An interesting application of CNN is extracting features, such as feature extraction from tumor images [16] or hyperspectral images [17]. It is observed that some simple shapes (e.g., edges and curves) can be extracted in shallow layers, and some semantic features can be extracted in a deeper layer [18]. In this paper, CNN is used to deal with the images generated by trimmed raw traffic and extract features from these images. The generation of images is discussed in section IV-B. In particular, 1D-CNN will be used; the specific reason will be mentioned in section IV-C.

IV. METHODOLOGY

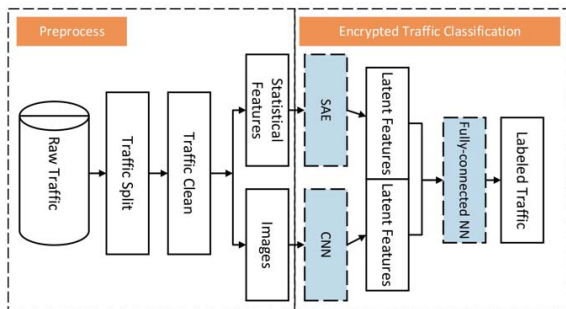


Figure 1. Overview of the structure of the proposed framework.

In this work, a new framework is proposed to classify encrypted traffic. As shown in “Figure. 1”, there are two

parts in this framework, data preprocessing and encrypted traffic classification. During the data preprocessing phase, we not only extract statistical features but also generate images from the trimmed raw traffic. The statistical features are used to compensate for information loss caused by trimming. During the classification phase, CNN and SAE are used at the same time. To be specific, the CNN is used to extract high-level features from the trimmed raw traffic, and the SAE is used to extract features from statistical features. Then we combine these two features into new high-level features and use these to classify encrypted traffic. The dataset, details of preprocessing, and the architecture of classification model are explained in the following.

TABLE I. 12 CLASSES IN “ISCX VPN-NON VPN” DATASET

Non-VPN/VPN	Traffic Type
non-VPN	Email
	Chat
	Streaming
	File Transfer
	P2P
	VoIP
VPN	VPN-Email
	VPN-Chat
	VPN-Streaming
	VPN-File Transfer
	VPN-P2P
	VPN-VoIP

A. Dataset

The experiments in this paper are carried out by using the “ISCX VPN-nonVPN” traffic dataset. This dataset was proposed by Draper-Gil et al. in 2016 [3]. Both regular (non-VPN) traffic and VPN traffic, captured from different applications (i.e., Skype, Gmail), are included in this dataset.

There are two types of data formats in this dataset, statistical features, and raw traffic. Because the correspondence between the statistical features and the raw traffic is not given in the dataset, we only use the raw traffic in this work. We extract statistical features from the raw traffic by ourselves. This will be introduced in section IV-B.

There are 12 classes in “ISCX VPN-non VPN”, including 6 classes of non-VPN traffic and 6 classes of VPN traffic. A detailed introduction of these 12 classes can be found in “Table I”.

B. Preprocess

During the phase of preprocessing, we convert the raw traffic to the expected format required by encrypted traffic classifier. There are three data preprocessing steps: traffic split, traffic clean, image, and statistical features generation. “Figure 2” shows the whole process of data preprocesses.

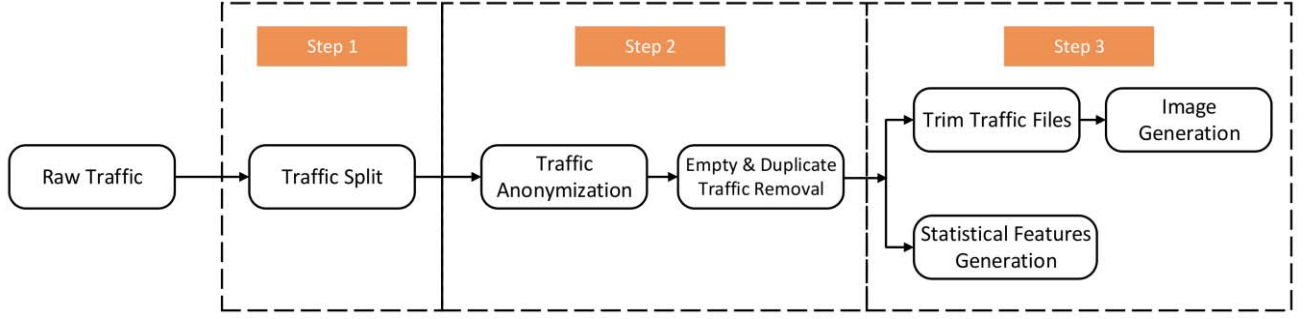


Figure 2. The process of data preprocess

TABLE II. LIST OF 26 STATISTICAL FEATURES

No.	Feature	Description
1	Num pkts	The number of packets in a session.
2	Avg syn flag	The average of packets with syn flag active in a session.
3	Avg urg flag	The average of packets with urg flag active in a session.
4	Avg fin flag	The average of packets with fin flag active in a session.
5	Avg ack flag	The average of packets with ack flag active in a session.
6	Avg psh flag	The average of packets with psh flag active in a session.
7	Avg rst flag	The average of packets with rst flag active in a session.
8	Avg DNS pkt	The average number of DNS packets in a session.
9	Avg TCP pkt	The average number of TCP packets in a session.
10	Avg UDP pkt	The average number of UDP packets in a session.
11	Avg ICMP pkt	The average number of ICMP packets in a session.
12	Duration window flow	The time from the first packet to last packet in a session.
13	Avg delta time	The average of delta times in a session. Delta time is the time from a packet to the next packet.
14	Min delta time	The minimum delta time in a session.
15	Max delta time	The maximum delta time in a session.
16	StDev delta time	The standard deviation of delta time in a session.
17	Avg pkts lenght	The average of packet lenghts in a session.
18	Min pkts lenght	The minimum of packet lenghts in a session.
19	Max pkts lenght	The maximum of packet lenghts in a session.
20	StDev pkts lenght	The standard deviation of packet lenghts in a session.
21	Avg small payload pkt	The average of packet with a small payload. A payload is considered small if his size is lower than 32 Byte.
22	Avg payload	The average of payload size in a session.
23	Min payload	The minimum of payload size in a session.
24	Max payload	The maximum of payload size in a session.
25	StDev payload	The standard deviation of payload size in a session.
26	Avg DNS over TCP	The average of ration DNS/TCP in a session.

Step 1 (Traffic Split). In this step, the continuous raw traffic is split into multiply discrete traffic units. There are two common ways to split the traffic, which are session and flow. A flow is a traffic unit having the same source IP, source port, destination IP, destination port, and transport-level protocol. Flow is one direction. The session is very similar to flow, but it includes both directions of flow [19]. Wang *et al.* tested these two ways in [4] and found that session is better than flow when used to classify encrypted traffic. Therefore, we choose to use the session in this work.

Step 2 (Traffic Clear). During the second step, we first anonymize the traffic by masking the MAC address and IP address, respectively. In the “ISCX VPN-nonVPN” dataset, each type of traffic has a unique IP address; therefore, the model may only use these features to classify the traffic.

Then we remove empty files, which do not have data in the application layer. The duplicated files are also removed in this step.

Step 3 (Image and Statistical Features Generation). Two actions are performed in this step, respectively. The first action is the generation of images. Because CNN needs a fixed-sized input, we trim all files to the same length (784 bytes in this paper). If the size of the file is larger than 784 bytes, then we trim it to 784 bytes. If the size of the file is shorter than 784 bytes, then we add 0x00 at the end of the file. The second action is to extract statistical features from the raw. Network traffic. These statistical features are used to supplement the session's structure information, such as the duration from the first packet to the last packet in a session.

26 statistical features are extracted in this paper. These features can be found in “Table II”.

After the above three steps, we obtained the images and statistical features corresponding to each session. These two

types of data are used in the following experiments, the images are the input of CNN, and the 26 statistical features are the input of SAE.

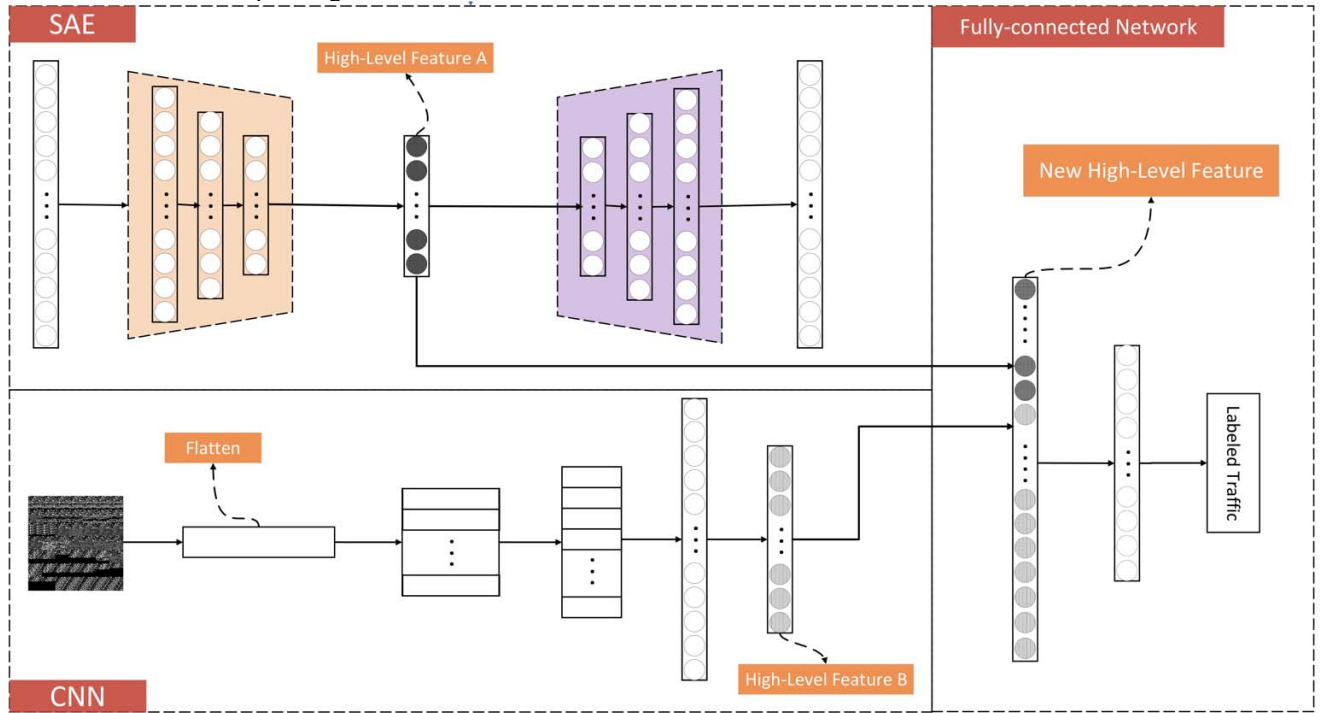


Figure 3. The architecture of the encrypted traffic classification model

C. Architecture

There are two types of neural networks in our proposed framework, namely SAE and CNN. “Figure. 3” shows detailed structure about the second part of the framework, which is an encrypted traffic classifier. This part includes two classes of input. First, the statistical features are used as the SAE’s input, and “high-level feature A” can be gotten through an encoder. Second, the images are used as the input of CNN. We use CNN to extract “high-level feature B”. After that, “high-level feature A” and “high-level feature B” are spliced as the “new high-level feature”. These new features include information both from statistical features and the trimmed raw traffic. We use the “new high-level feature” to classify encrypted network traffic.

The main parameters of the models in the second part of our framework are described in “Table III”. We use 1D-CNN in our model because network traffic is sequential data. Additionally, Wang et al. [4] has used an experiment to prove that when using the raw traffic for classification, 1D-CNN is better than 2D-CNN. 1D-CNN’s parameters in our work refer to Wang’s [4]. The input of CNN is 784 bytes, the input of SAE is 26 bytes, and the ratio of these two inputs is 30:1. Therefore, we keep this ratio in the “new high-level feature” when we combine the output from two deep learning models. Specifically, the output of CNN is a 300 bytes vector, and the output of the encoder is a 10 bytes vector. Finally, the new vector is a 310 bytes vector.

TABLE III. MAIN PARAMETERS OF THE MODEL

Layer	Operation	Input	Filter	Output
CNN-1	conv+ReLU	1*784	1*25	32*784
CNN-2	1d max pool	32*784	1*3	32*262
CNN-3	conv+ReLU	32*262	1*25	64*262
CNN-4	1d max pool	64*262	1*3	64*88
CNN-5	flatten	64*88	-	5632
CNN-6	fully-connected	5632	-	1024
CNN-7	fully-connected	1024	-	300
SAE-Encode-1	fully-connected	26	-	18
SAE-Encode-2	fully-connected	18	-	10
SAE-Decode-3	fully-connected	10	-	18
SAE-Decode-4	fully-connected	18	-	26
Fully-connected-1	fully-connected	300+10	-	100
Fully-connected-2	fully-connected	100	-	30
Fully-connected-3	softmax	30	-	12

V. EXPERIMENTS AND RESULTS

A. Experiment Setup

All the experiments are conducted on Google Colab. Pytorch [20] is used to build the proposed neural networks. We randomly take 10% of the data as the test data, and the rest is the training data. The batch size is 200. The CNN and SAE are trained by Adam optimizer [21]. The initial learning

rate is 0.001 and decay of 0.9 per 20 epochs. The total training time is 100 epochs. Before training the whole model, we first pre-train the SAE to find good initial weights.

B. Evaluation Metrics

Four evaluation metrics are used to evaluate the performance of the classifier: accuracy, precision, recall, and F1-score. These evaluation metrics can be calculated in the following ways.

1) **Accuracy**: The ratio of correctly recognized traffic to the entire test dataset ($\text{Accuracy} \in [0,1]$).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

2) **Precision**: The ratio of the number correctly classified as specific type traffic to the number of all samples classified as this type ($\text{Precision} \in [0,1]$).

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

3) **Recall**: The ratio of the number correctly classified as a specific type of traffic to all this type of traffic ($\text{Recall} \in [0,1]$).

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

4) **F1-score**: F1-score is used to measure precision and recall at the same time. It uses the harmonic mean in place of the arithmetic mean ($\text{F1-score} \in [0,1]$).

$$\text{F1-score} = \frac{2 * \text{Recall} * \text{Precision}}{\text{Recall} + \text{Precision}} \quad (6)$$

where TP, TN, FP, FN stand for true positive, true negative, false positive, and false negative, respectively.

C. Experiment Results and Analysis

“Table IV” shows the experiment results of each type of encrypted traffic. We compare the precision, recall, and f1-score obtained by our framework with these metrics from [10]. It can be seen that except for the classification performance on File Transfer, our proposed framework’s performances on other types of traffic are much better. The weighted average F1-score achieved by our framework is 0.98, 0.05 larger than that in [10]. This is because our framework uses both raw traffic and statistical features. Specifically, we use the CNN to extract features from the trimmed raw traffic and use the encoder in SAE to compress the statistical features. Then we combine the outputs from the above two models into new high-level features. These new features, which include information both from trimmed raw traffic and statistical features, are used to classify encrypted traffic.

To further evaluate the performance of the proposed framework, the results of our framework are compared with other state-of-the-art models. “Table V” shows the performance comparison on the “ISCX VPN-non VPN”

traffic dataset between our proposed framework and others. It can be seen from “Table V”, compared with the other four models, our model achieves the highest overall precision, recall, F1-score, and accuracy.

Finally, we explore the property of these new features, whether they can represent different types of traffic. In order to demonstrate the “new high-level feature” that can represent different classes of traffic, t-SNE [22] is used for dimension reduction and visualization. The t-SNE is suitable for mapping high-dimensional data to a low-dimensional space for visualization. As shown in “Figure. 4”, the “new high-level feature” are projected into 2D space. Different types of traffic are represented by different colors and different shapes in “Figure. 4”. It can be seen that the same types of traffic cluster in similar positions, while the distance between different types of traffic is large. This shows that the “new high-level feature” in our framework can represent the characteristics of different types of traffic well.

TABLE IV. PERFORMANCE FOR EACH TYPE OF ENCRYPTED TRAFFIC CLASSIFICATION

Traffic Class	The Proposed Framework			CNN[8]		
	Pr	Re	F ₁	Pr	Re	F ₁
Chat	0.99	1.00	1.00	0.71	0.84	0.77
Email	1.00	1.00	1.00	0.87	0.96	0.91
File Transfer	0.91	0.91	0.91	1.00	0.98	0.99
P2P	0.99	1.00	1.00	1.00	1.00	1.00
Streaming	0.96	0.99	0.98	0.87	0.92	0.90
VoIP	0.98	0.97	0.97	0.88	0.63	0.74
VPN-Chat	1.00	0.99	0.99	0.98	0.98	0.98
VPN-Email	0.97	1.00	0.98	0.98	0.99	0.99
VPN-File Transfer	0.99	0.98	0.98	0.99	0.99	0.99
VPN-P2P	0.96	0.98	0.97	1.00	1.00	1.00
VPN-Streaming	1.00	1.00	1.00	1.00	1.00	1.00
VPN-VoIP	1.00	1.00	1.00	1.00	0.99	1.00
Weighted avg	0.98	0.98	0.98	0.94	0.93	0.93

TABLE V. A COMPRASION BETWEEN DIFFERENT MODELS

Classifier	Precision	Recall	F1-score	Accuracy
Deep packet-CNN [8]	0.94	0.93	0.93	-
Deep packet-SAE [8]	0.92	0.92	0.92	-
C4.5 [3]	-	-	-	0.90
FlowPic [20]	-	-	-	0.88
The proposed framework	0.98	0.98	0.98	0.98

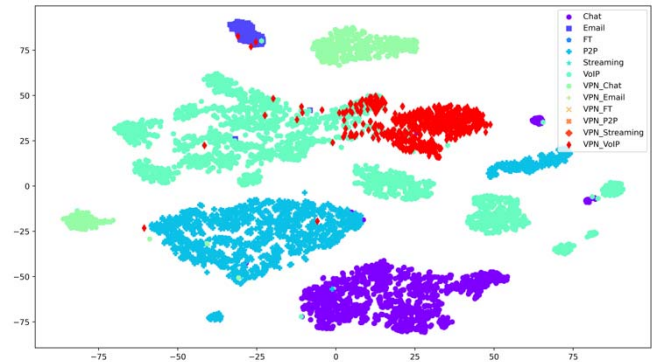


Figure 4. Visualizations of 12 types of encrypted traffic

VI. CONCLUSION AND FUTURE WORK

Nowadays, deep learning-based encrypted traffic classification is becoming more and more popular. However, the input of the neural networks needs fixed-sized data, so that the raw traffic needs to be trimmed first before classification, which will cause information loss. Therefore, a new framework is proposed in this paper, which utilizes both the raw traffic and the statistical features of traffic.

There are two parts of this framework, data preprocessing and encrypted traffic classification. During the phase of data preprocessing, we extract the statistical features from the raw traffic and then trim the raw network traffic to make it suitable for CNN's input. During the phase of classification, we use a CNN to extract high-level features from raw network traffic and use an SAE to encode the statistical features. Then we combine these two kinds of features to a new high-level feature, which includes the information both from the trimmed raw traffic and statistical features. Finally, this new high-level feature is used to classify encrypted traffic. The result shows that our framework gets a higher f1-score on encrypted traffic classification; the f1-score is 0.98. In addition, we visualize the new high-level features and find that these features can represent different classes of traffic well.

The present work still has room for improvement. Firstly, we only use one dataset, "ISCX VPN-nonVPN", to verify our proposed framework in this work. More datasets should be used to check the performance of our framework and whether the new high-level features are representative. Secondly, statistical features need to be extracted in our proposed framework; it will be more time-consuming than only using deep neural networks to extract features from network traffic.

Notwithstanding these limitations, this study offers valuable insight into using convolutional neural networks and stacked autoencoders at the same time to classify the encrypted traffic. Additionally, we generate new high-level features that contain both information from trimmed raw traffic and statistical features, representing different classes of traffic well. In the future, more experiments can focus on experimenting on more datasets, making the framework work in real-time, and makes this framework more robust.

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